

PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

ServiceNow Project Documentation

Team ID: SWTID-2026-2467

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Maximum Marks:

1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification using Flow Designer is a ServiceNow automation project that classifies common school/IT support tickets into predefined Category and Subcategory values. The solution is implemented using native ServiceNow configuration and Flow Designer rather than an external application.

The project uses an Incident Workflow custom table and form to capture ticket information. When a record is created with Category blank, the classification flow evaluates the ticket information and applies the appropriate classification using configured conditional branches and Update Record actions.

Issue Type	Category	Subcategory
Wi-Fi	Network	Wi-Fi
Projector	Hardware	Projector
Forgot Password	Access	Forgot Password
Slow Computer	Performance	Slow Computer

The Subcategory field is configured as dependent on Category, helping keep the classification values consistent.

1.2 Purpose

- Reduce repetitive manual classification of IT support tickets.
- Provide consistent classification through predefined business rules.
- Improve the quality and completeness of Incident Workflow records.
- Reduce incorrect Category/Subcategory combinations.
- Provide a maintainable automation that can be extended with additional rules.
- Demonstrate practical ServiceNow development using tables, forms, fields, Flow Designer, access controls and Update Sets.

2. IDEATION PHASE

2.1 Problem Statement

IT support tickets are often created by users who know the issue but may not know the correct service-management category. Manual classification can result in inconsistent categories, invalid subcategories or incomplete ticket information. These issues create additional review work for support personnel.

The project therefore addresses the following problem: How can common IT support tickets be classified automatically and consistently within ServiceNow while keeping the solution simple enough for a project team to configure, test and maintain?

2.2 Empathy Map Canvas

Area	Observed User Need / Behaviour
Says	"I need to report my issue quickly." "I want the ticket to be categorized correctly."
Thinks	The correct category may not always be obvious when creating a ticket.
Does	Creates a ticket, provides issue details, saves the record and waits for support.
Feels	May feel frustrated when a ticket is categorized incorrectly or requires clarification.
Pain Points	Manual classification, inconsistent selections and additional correction effort.
Needs	Simple ticket entry, consistent classification and controlled subcategory choices.
Gains	Less manual classification, cleaner records and a more predictable support workflow.

2.3 Brainstorming

- Use ServiceNow Flow Designer because the platform already manages the ticket record.
- Trigger classification when a newly created ticket requires classification.
- Create a controlled Category/Subcategory mapping for frequent IT issues.
- Configure Subcategory as dependent on Category.
- Use If / Else If branches for the supported issue types.
- Use Update Record actions to write the resulting classification.
- Use Update Sets to capture configuration changes.
- Validate the implementation with functional and UAT test cases.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Activity	Pain Point	System Response	Benefit
Identify issue	Recognize Wi-Fi, projector, password or computer issue.	Correct classification may be unclear.	Structured Incident Workflow form.	Standardized capture.
Create ticket	Enter caller and issue details.	Manual classification may be inconsistent.	Record stores ticket details.	Complete ticket record.
Classify	Ticket needs Category/Subcategory.	Wrong classification can delay handling.	Flow evaluates configured conditions.	Consistent classification.
Update record	Classification is written to ticket.	Manual correction otherwise needed.	Update Record action updates fields.	Reduced manual effort.
Support handling	Support team processes ticket.	Misclassification can affect handling.	Structured category supports processing.	Improved consistency.
Close / retain	Ticket remains as historical record.	Poor data quality reduces usefulness.	Classification remains on record.	Better operational data.

3.2 Solution Requirements

Type	Requirement
Functional	Create and maintain Incident Workflow table and form.
Functional	Capture Caller, Category, Subcategory, Short Description, Description and State.
Functional	Configure supported Category and Subcategory values.
Functional	Make Subcategory dependent on Category.
Functional	Trigger Flow Designer for records requiring classification.
Functional	Evaluate supported ticket conditions.
Functional	Update the ticket with the resulting classification.
Security	Use configured access controls/roles for the custom table.
Deployment	Capture project configuration in an Update Set.
Testing	Verify classification and configuration through functional/UAT scenarios.

3.3 Data Flow Diagram

The diagram below presents the project flow from ticket creation to automatic classification and record update. It is intentionally separated into clear stages so that the data movement can be understood without relying on a crowded screenshot.

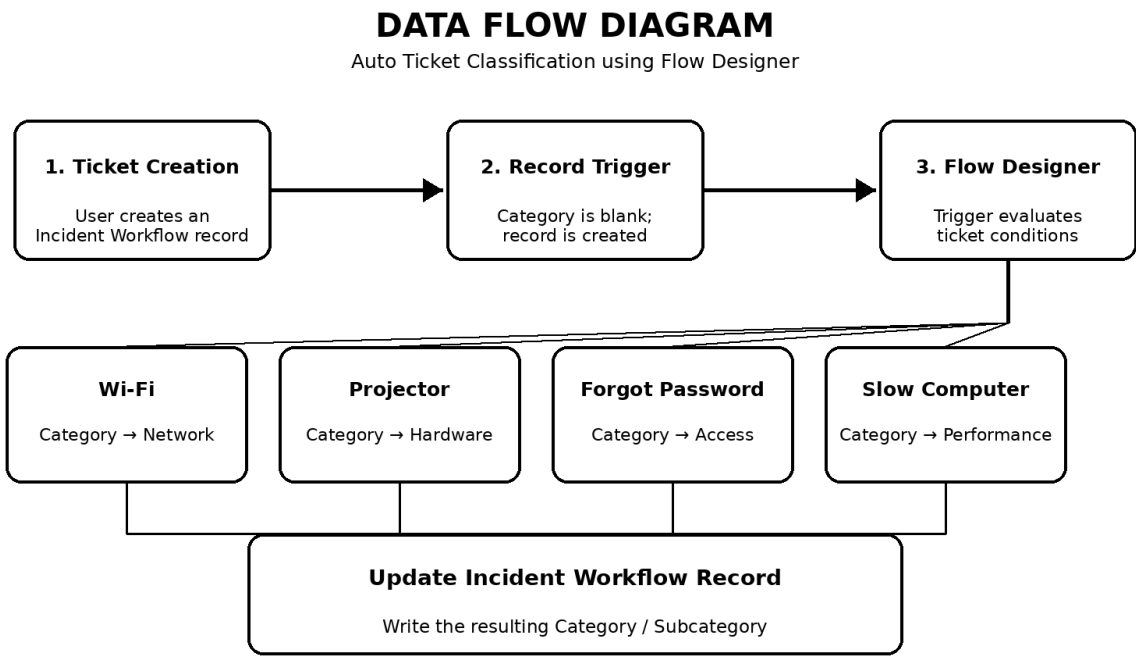
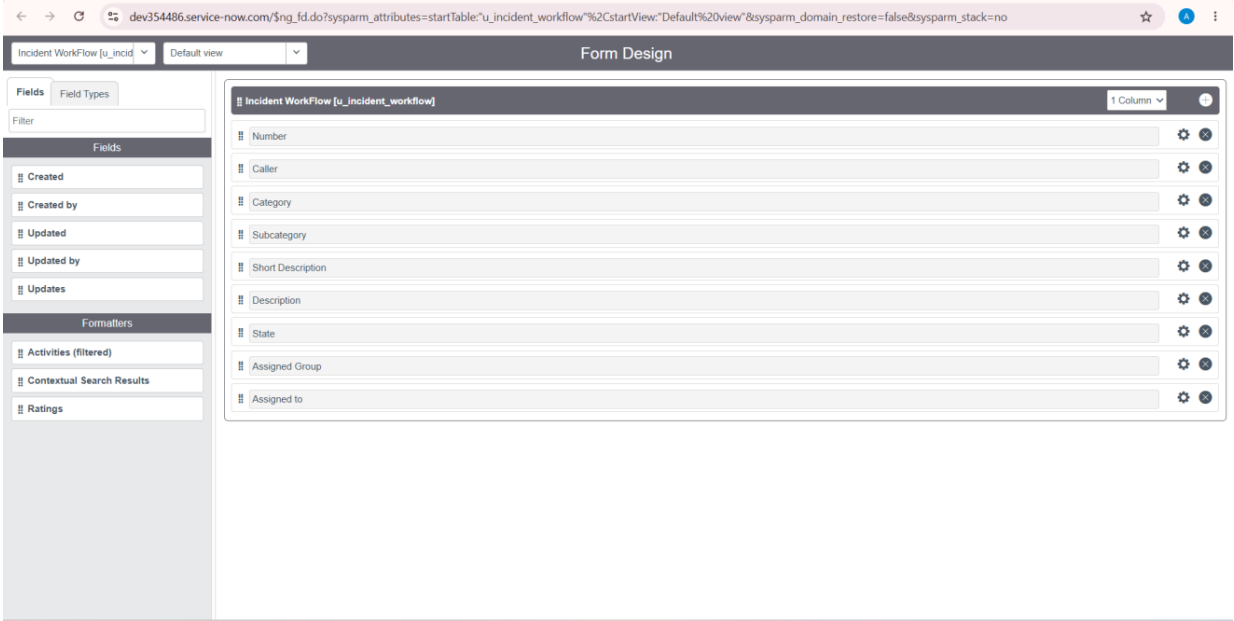


Figure 1. Clean Data Flow Diagram – Auto Ticket Classification.

3.4 Technology Stack

Component	Role in Project
ServiceNow Platform	Development and execution environment.
Incident Workflow Custom Table	Stores project ticket records.
ServiceNow Form / Form Designer	Provides the user-facing ticket structure.
Flow Designer	Implements trigger, conditions and record updates.
Category / Subcategory Fields	Store classification information.
Dependent Field Configuration	Controls Subcategory according to Category.
Access Controls	Provide controlled access to the custom table.
Update Set	Captures project configuration for transfer/deployment.
Web Browser	Used to access the ServiceNow Developer Instance.



The screenshot displays the ServiceNow Form Designer interface for the Incident Workflow form. The interface is divided into two main sections: a left sidebar and a main configuration area. The left sidebar contains a list of fields and formatters. The main configuration area shows the configuration for the Incident Workflow form, including the field types and formats for each field.

Fields:

- Created
- Created by
- Updated
- Updated by
- Updates

Formatters:

- Activities (filtered)
- Contextual Search Results
- Ratings

Incident Workflow [u_incident_workflow] Fields:

- Number
- Caller
- Category
- Subcategory
- Short Description
- Description
- State
- Assigned Group
- Assigned to

Figure 2. Incident Workflow form configuration showing the project fields.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution	Expected Fit
Manual classification	Flow Designer applies predefined rules.	Reduces repetitive manual work.
Incorrect category	Configured mappings define expected classification.	Improves consistency.
Invalid subcategory	Subcategory depends on Category.	Controls valid combinations.
Blank classification	Flow trigger checks records with Category blank.	Targets records requiring classification.
Configuration transfer	Update Set captures project changes.	Supports controlled deployment.
Need for verification	Functional/UAT cases test supported scenarios.	Provides documented evidence.

4.2 Proposed Solution

- Create the Incident Workflow table and configure its form fields.
- Configure Category and Subcategory values for supported IT issues.
- Configure Subcategory dependency on Category.
- Create the Auto Classify School IT Tickets flow in Flow Designer.
- Use the record-created trigger with Category blank.
- Evaluate the supported issue conditions using If / Else If branches.
- Use Update Record actions to write the classification.
- Verify the configuration through functional and UAT testing.
- Capture the final project configuration in the Project Update Set.

4.3 Solution Architecture

The architecture separates the user-facing record, automation logic, stored configuration and deployment/security controls. This provides a clearer view of how the ServiceNow components work together.

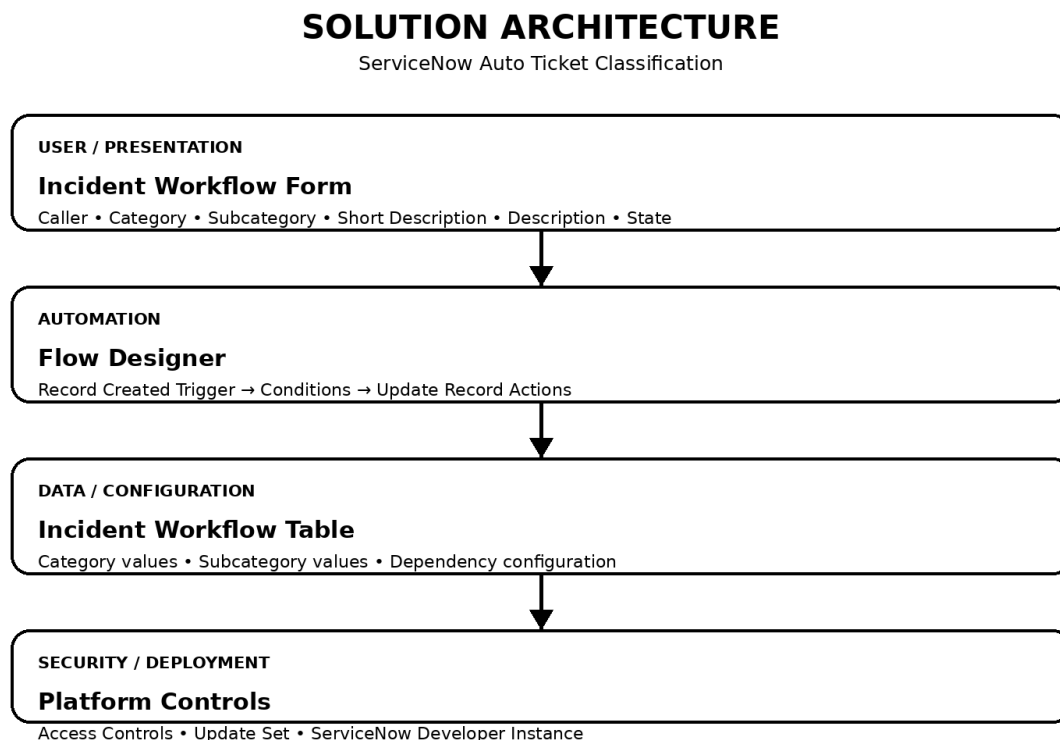


Figure 3. Clean Solution Architecture – ServiceNow implementation.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is planned using the Sprint → Epic → User Story → Story Point structure from the supplied planning reference. Two sprints are used to organize development, automation and testing.

Sprint	Epic	User Story	Points
Sprint 1	Incident Workflow Setup	Create Incident Workflow Table (USN1)	3
Sprint 1	Incident Workflow Setup	Configure Incident Workflow Form (USN2)	3
Sprint 1	Incident Workflow Setup	Configure Access Controls (USN3)	2
Sprint 1	Category & Subcategory Configuration	Create Category Values (USN4)	2
Sprint 1	Category & Subcategory Configuration	Create Subcategory Values (USN5)	2
Sprint 1	Category & Subcategory Configuration	Configure Subcategory Dependency (USN6)	3
Sprint 2	Auto Classification Flow	Create Flow Designer Trigger (USN7)	2
Sprint 2	Auto Classification Flow	Configure Classification Conditions (USN8)	3
Sprint 2	Auto Classification Flow	Configure Update Record Actions (USN9)	3
Sprint 2	Testing & Deployment	Execute Functional and UAT Scenarios (USN10)	3
Sprint 2	Testing & Deployment	Validate Classification Mappings (USN11)	2
Sprint 2	Testing & Deployment	Complete Update Set and Submission (USN12)	2

Sprint 1 Total = 15 Story Points.

Sprint 2 Total = 15 Story Points.

Total Story Points = 30.

Number of Sprints = 2.

Velocity = $30 / 2 = 15$ Story Points per Sprint.

Planning Milestones

Milestone	Completion Evidence
Ticket structure	Incident Workflow table/form configured.
Classification configuration	Category/Subcategory mapping and dependency configured.
Automation	Flow Designer classification branches and Update Record actions configured.
Testing	Functional and UAT scenarios documented.
Deployment readiness	Project Update Set shown in Complete state.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

This project is a rule-based ServiceNow Flow Designer automation, not a trained machine-learning model. Therefore, training accuracy, validation accuracy and object-detection confidence scores are not appropriate measures for this implementation. Performance testing in this report focuses on functional behavior and configuration-level readiness.

ID	Scenario	Test Steps	Expected Result	Result
FT-01	Wi-Fi ticket classification	Create ticket → provide Wi-Fi issue → process	Category = Network	Pass
FT-02	Projector ticket classification	Create ticket → provide Projector issue → process	Category = Hardware	Pass
FT-03	Forgot Password classification	Create ticket → provide Forgot Password issue → process	Category = Access	Pass
FT-04	Slow Computer classification	Create ticket → provide Slow Computer issue → process	Category = Performance	Pass
FT-05	Dependent Subcategory	Select Category → inspect Subcategory	Valid values depend on Category	Pass
FT-06	Blank Category trigger	Create record with Category blank → process	Flow evaluates configured branches	Pass

Test Evidence

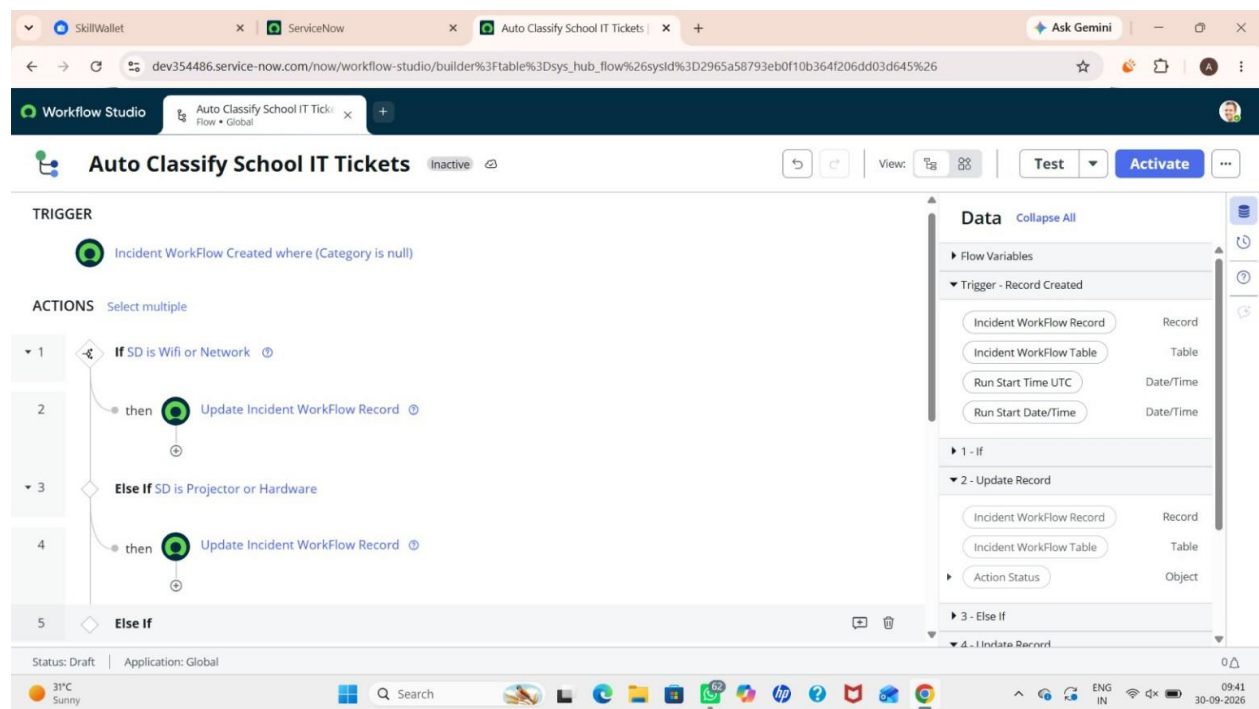


Figure 4. Flow Designer main classification logic. The screenshot is used as logic/configuration evidence; its displayed Draft/Inactive state is not claimed as proof of runtime execution.

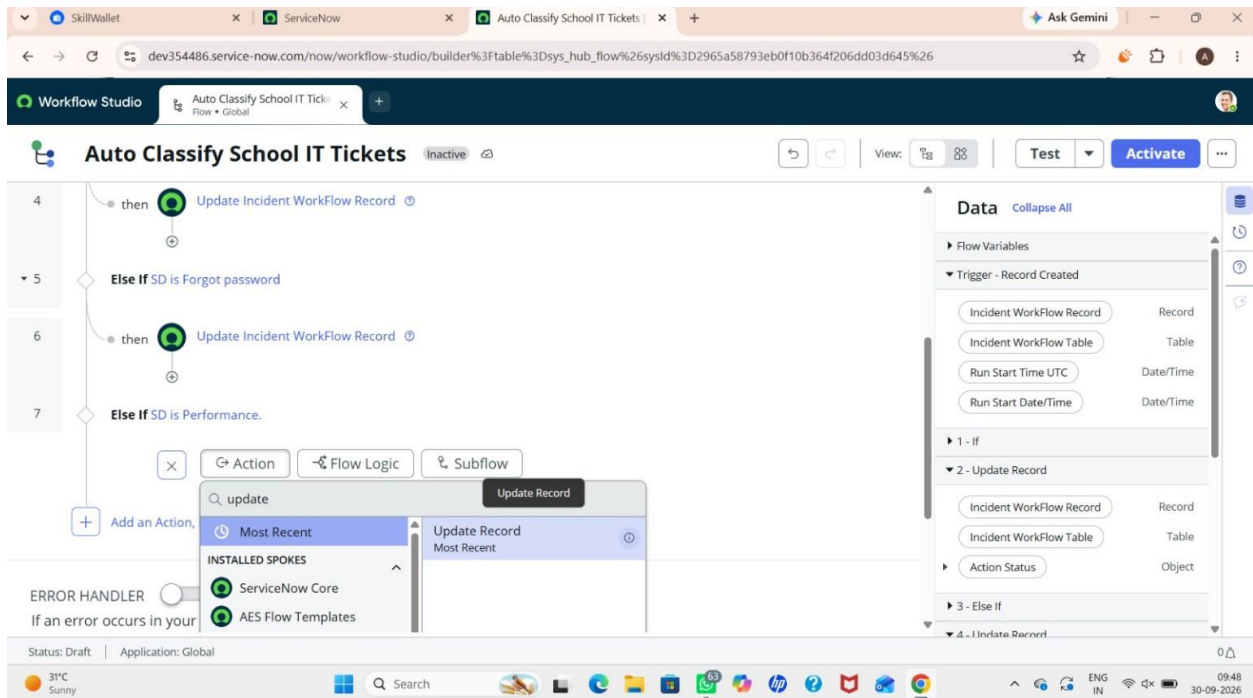


Figure 5. Flow Designer continuation showing Forgot Password / Performance branches and Update Record action selection.

7. RESULTS

7.1 Output Screenshots

The results section contains only screenshots that directly support the implemented ServiceNow configuration or project output. Screenshots are grouped by what they demonstrate rather than inserted merely to increase page count.

A. Ticket Form and Configuration

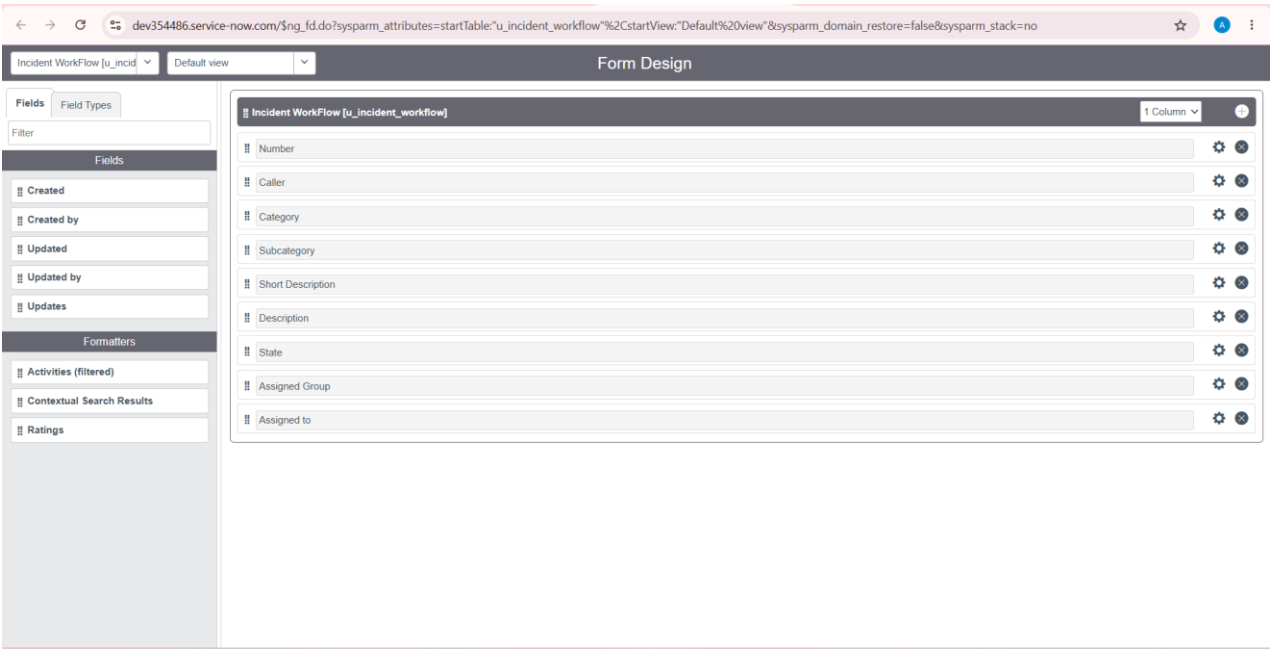


Figure 6. Incident Workflow Form Designer showing Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned to.

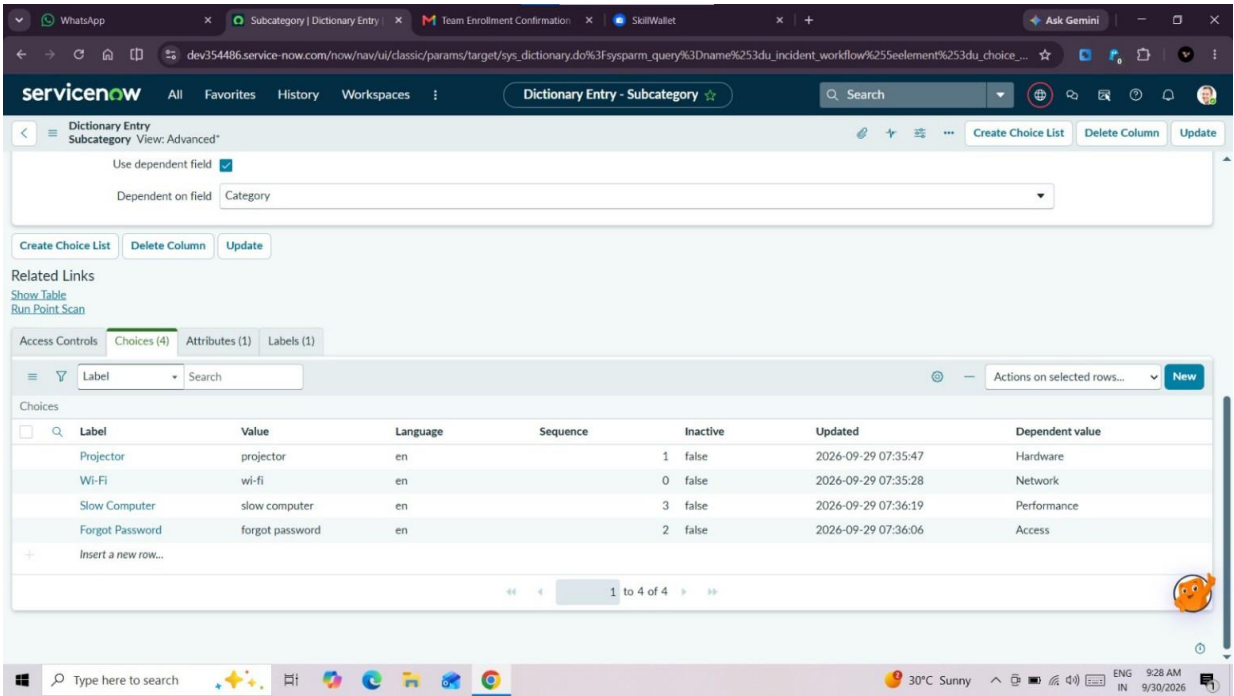


Figure 7. Category/Subcategory mapping configuration.

B. Dependent Field Configuration

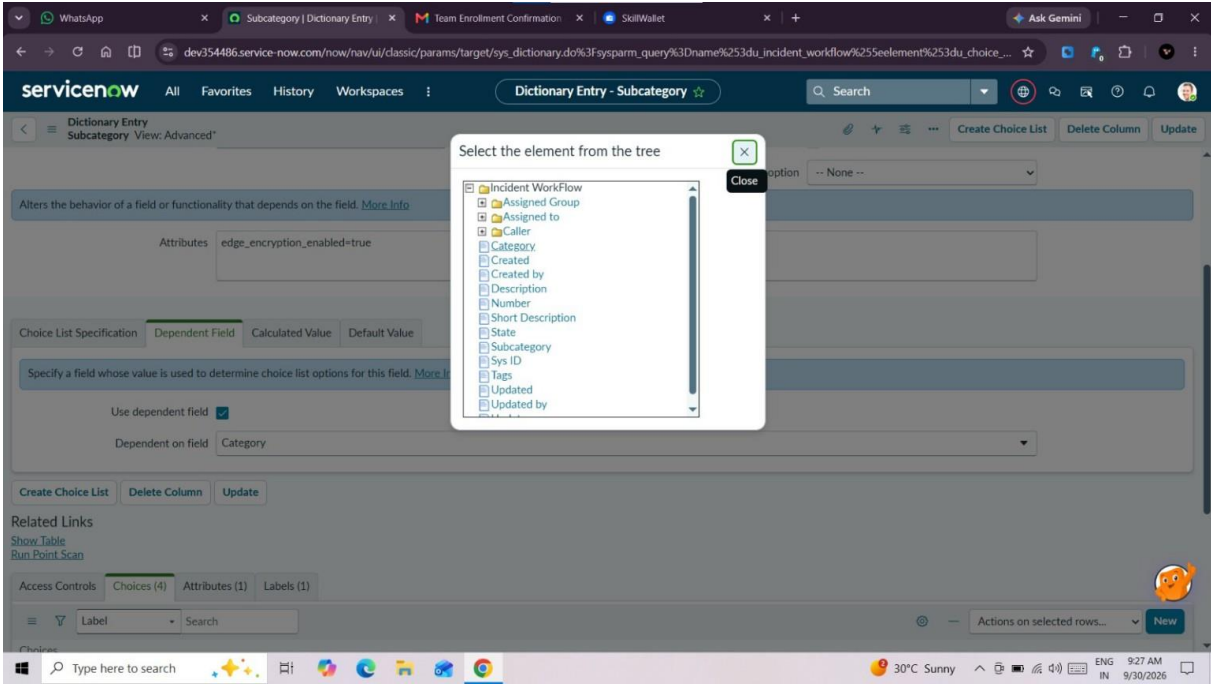


Figure 8. Category/Subcategory dependency configuration.

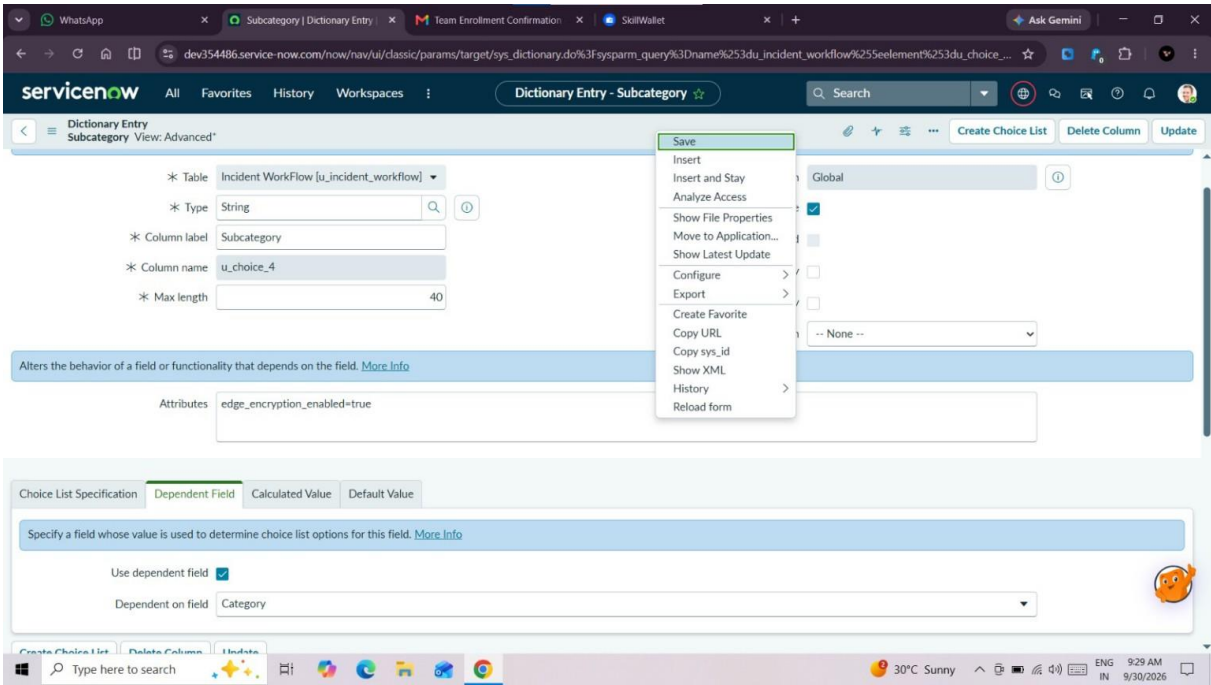


Figure 9. Subcategory field configuration showing its dependency on Category.

C. Classification Flow

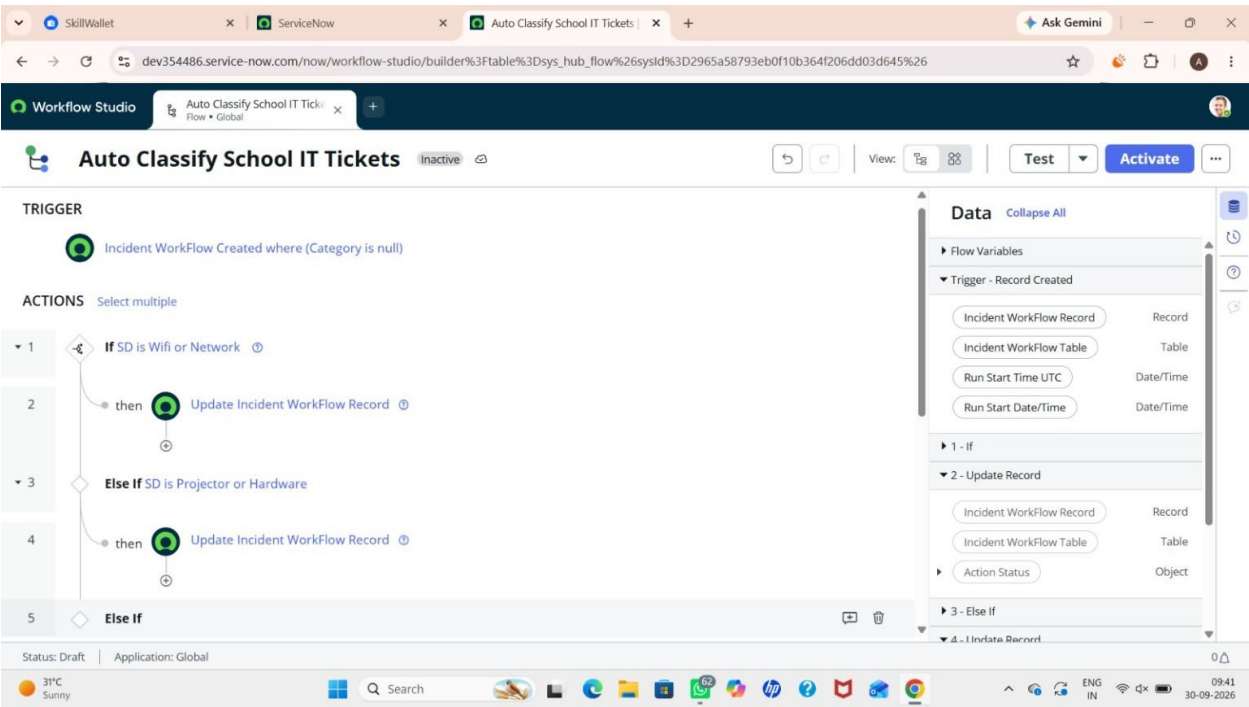


Figure 10. Main Flow Designer classification branches.

D. Update Action and Ticket Confirmation

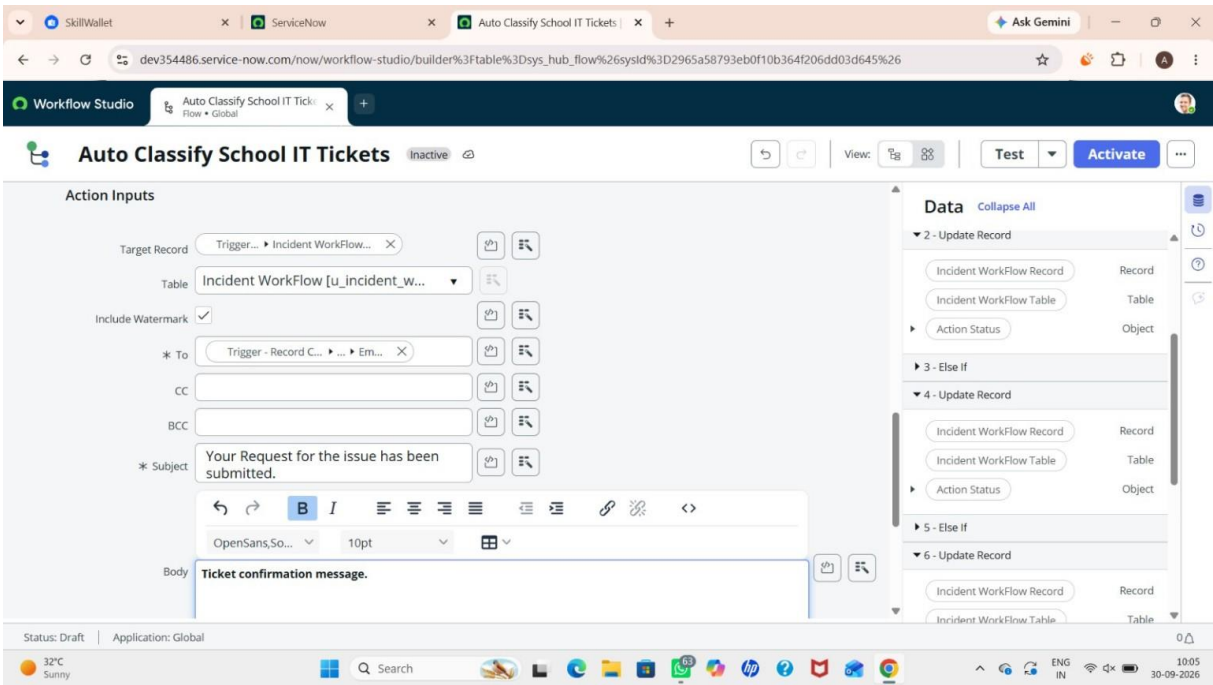


Figure 11. Flow Designer Update Record action / supporting configuration.

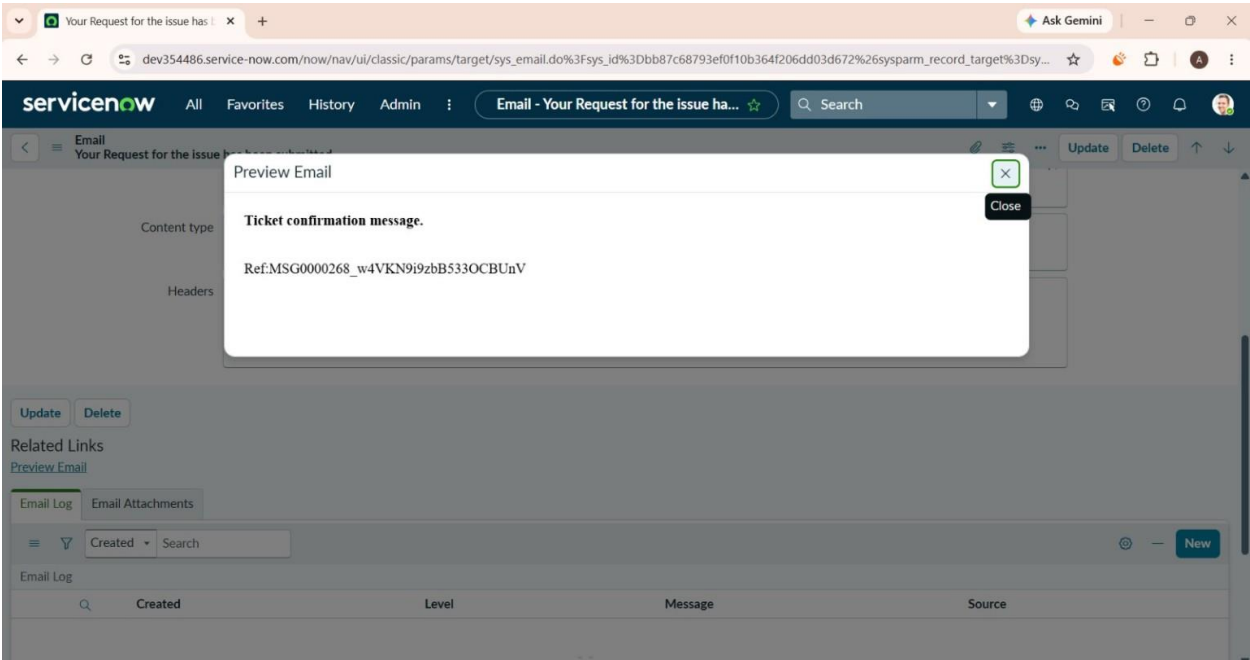


Figure 12. Ticket confirmation email preview. This is included as notification/output evidence, not as proof of classification accuracy.

E. Deployment / Configuration Capture

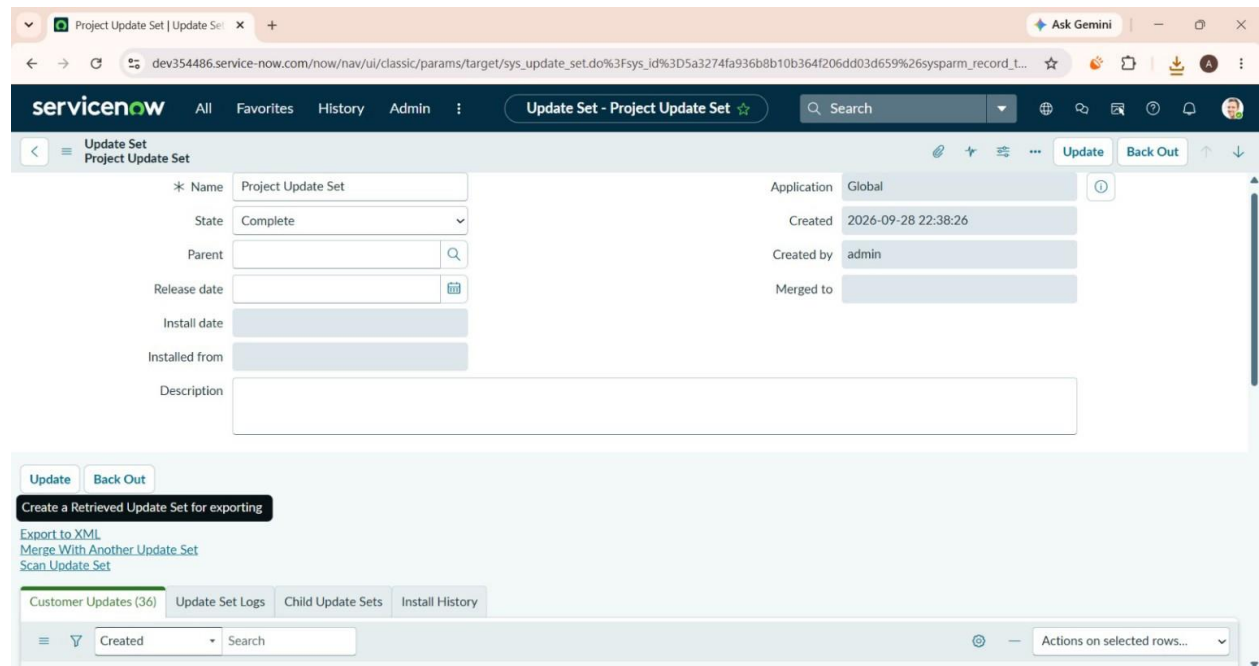


Figure 13. Project Update Set in Complete state with customer updates visible.

Classification Evidence Summary

Issue	Expected Category	Configured Evidence
Wi-Fi	Network	Category/Subcategory mapping and Flow condition.
Projector	Hardware	Category/Subcategory mapping and Flow condition.
Forgot Password	Access	Category/Subcategory mapping and Flow condition.
Slow Computer	Performance	Category/Subcategory mapping and Flow condition.

The four mappings above are the core classification outcomes documented by the project configuration. The ticket screenshots are treated as supporting configuration evidence and are not presented as independent accuracy measurements.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces repetitive manual classification effort.
- Provides consistent classification for supported issue types.
- Uses native ServiceNow Flow Designer functionality.
- Maintains controlled Category/Subcategory relationships.
- Can be extended with additional conditions.
- Does not require an external application for the core workflow.
- Update Set provides a mechanism to capture project configuration.
- The logic is visible and reviewable within Flow Designer.

Disadvantages

- The current flow handles only the issue patterns explicitly configured.
- Additional issue types require new conditions and testing.
- Rule-based logic may not interpret unexpected free-text descriptions.
- The current project does not use machine learning or natural-language classification.
- Production-scale response-time/load testing was not part of the available evidence.
- Correct operation depends on ServiceNow configuration and access permissions.

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project demonstrates a complete ServiceNow workflow automation for common IT support-ticket classification. The implementation combines a custom Incident Workflow table and form, controlled Category/Subcategory values, dependent-field configuration, Flow Designer conditions, Update Record actions, access control configuration and an Update Set.

The implemented mapping covers Wi-Fi → Network, Projector → Hardware, Forgot Password → Access and Slow Computer → Performance. The flow structure provides a clear, maintainable rule-based approach for these supported scenarios.

The project therefore demonstrates the practical use of ServiceNow configuration and Flow Designer to reduce manual classification effort and improve consistency in ticket records.

10. FUTURE SCOPE

- Add more IT categories such as printer, VPN, email, software installation and account issues.
- Use Natural Language Understanding or machine learning to classify free-text ticket descriptions.
- Automatically assign tickets to support groups based on classification.
- Add priority and impact recommendations.
- Add notifications and escalation workflows.
- Create ServiceNow reporting dashboards for category trends and ticket volume.
- Add automated regression testing when new classification rules are introduced.
- Measure production response time and throughput after deployment if operational targets are defined.
- Add audit reporting for classification changes.

11. APPENDIX

GitHub & Project Demo Link

GitHub Repository: <https://github.com/abinaya1947/Auto-Ticket-Classification-using-Flow-Designer.git>

Project Demo: https://drive.google.com/file/d/1Q5NON8X0MTCsh-sc-w3ZZWlyRVXi38N4/view?usp=drive_link

Project Evidence Index

Figure	Evidence	Purpose
1	Data Flow Diagram	Explains movement from ticket creation to classification and record update.
2	Incident Workflow Form	Shows configured project fields.
3	Solution Architecture	Shows the ServiceNow component layers.
4-5	Flow Designer Screens	Shows classification conditions and Update Record actions.
6-7	Form and Mapping	Shows ticket structure and category mappings.
8-9	Dependency Configuration	Shows Subcategory dependency on Category.
10-11	Flow / Update Action	Shows automation logic and record update configuration.
12	Email Preview	Shows ticket confirmation notification output.
13	Project Update Set	Shows configuration captured in Complete state.